RAJALAKSHMI ENGINEERING COLLEGE

Department of Artificial Intelligence and Machine Learning

AD23632 – Framework for Data and Visual Analytics (FDVA)

MINI PROJECT REPORT

FINANCIAL DATA ANALYTICS

Submitted in partial fulfilment of the requirements for the course
Framework for Data and Visual Analytics

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TABLE OF CONTENTS

Abstract 3

Chapter 1: Introduction 4

Chapter 2: Tools and Technologies Used 6

Chapter 3: Dataset Description 7

Chapter 4: Methodology 9

Chapter 5: Power BI Dashboard and Visualizations 11

Chapter 6: Results and Discussion 14

Chapter 7: Advantages and Limitations 15

Chapter 8: Power BI Dashboard Output 16

Chapter 9: Conclusion 17

ABSTRACT

Organizations record large volumes of financial transactions every day, including orders, customer purchases, product sales, and revenue. Analyzing this data helps businesses understand revenue sources, identify their most valuable customer segments and regions, and recognize seasonal patterns that support planning and decision-making.

The project titled Financial Data Analytics focuses on analyzing sales transaction data using Python and Microsoft Power BI. Python is used for data inspection, cleaning, and preprocessing. Power BI is used to build an interactive dashboard that presents key financial metrics and trends in an understandable visual format.

The dashboard includes key performance indicators such as total sales, total orders, total customers, and total products. It also presents the distribution of customer records by region, sales by customer segment, and the monthly sales trend over the year.

The analysis shows total sales of approximately 2.26 million from 4,922 orders placed by 793 customers across 1,861 products. The Consumer segment contributes the largest share of sales, the West region has the largest share of records, and sales rise strongly in the last four months of the year.

The project demonstrates how data analytics and visualization can convert raw transaction data into meaningful financial insights.

Keywords: Financial Analytics, Sales Analysis, Python, Power BI, Data Visualization, KPI Dashboard, Exploratory Data Analysis.

CHAPTER 1: INTRODUCTION

1.1 Introduction

Financial and sales data is one of the most valuable assets of any business. Every order placed by a customer records information such as the product purchased, the customer's segment, the region of purchase, the date, and the sales value. When this data grows into thousands of records, reviewing it manually becomes impractical.

Financial Data Analytics applies data analysis techniques to such records to measure revenue, understand customer behavior, and identify trends. Data visualization tools present the results through KPI cards, charts, and dashboards so that decision-makers can interpret them quickly.

This project uses Python for data preprocessing and exploratory analysis and Microsoft Power BI for building a two-page dashboard. The dashboard enables users to view overall sales performance, compare customer segments and regions, and study how sales change month by month.

1.2 Problem Statement

Sales transaction data contains many fields and thousands of records. Examining the records individually makes it difficult to calculate total revenue, compare segments and regions, or recognize seasonal patterns.

Therefore, an analytical solution is required to process the transaction data, calculate important financial metrics, and present the results through clear visualizations.

1.3 Objectives

The main objectives of the project are:

1. To analyze sales transaction data using Python.
2. To preprocess and clean the dataset.
3. To calculate total sales, total orders, total customers, and total products.
4. To compare sales across customer segments.
5. To analyze the distribution of customer records across regions.
6. To examine the monthly sales trend.
7. To develop a Power BI dashboard for presenting analytical insights.

1.4 Scope of the Project

The project focuses on analyzing a sample sales dataset containing order, customer, product, segment, region, and sales information.

The scope includes:

- Data cleaning and preprocessing.
- Exploratory data analysis using Python.
- Calculation of key financial metrics.
- Comparison of customer segments and regions.
- Analysis of monthly sales trends.
- Development of a multi-page Power BI dashboard.

The analysis is limited to the data available in the selected dataset. The results do not necessarily represent the current performance of any real business.

CHAPTER 2: TOOLS AND TECHNOLOGIES USED

2.1 Python

Python is a high-level programming language widely used for data analysis and visualization. In this project, Python is used to inspect the dataset, identify missing values and duplicate records, convert date information, and perform exploratory analysis.

Libraries Used

- Pandas: Used for loading, cleaning, manipulating, and analyzing tabular data.
- NumPy: Used for numerical operations and calculations.
- Matplotlib: Used to create charts and graphs during exploratory analysis.

2.2 Microsoft Power BI

Microsoft Power BI is a business intelligence and data visualization tool used to transform data into interactive reports and dashboards.

In this project, Power BI is used to:

- Import the sales dataset.
- Create aggregate measures such as sum and count.
- Display key performance indicators using KPI cards.
- Visualize customer distribution by region using a donut chart.
- Compare sales by segment using a column chart.
- Present the monthly sales trend using a line chart.

2.3 CSV Dataset

The dataset is stored in Comma-Separated Values (CSV) format. This format stores tabular data in rows and columns and can be imported into both Python and Power BI.

2.4 Technologies Summary

Table 2.1: Technologies and Their Purpose

Technology	Purpose
Python	Data preprocessing and analysis
Pandas	Data manipulation
NumPy	Numerical operations
Matplotlib	Exploratory visualization
Microsoft Power BI	Interactive dashboard development
CSV	Dataset storage

CHAPTER 3: DATASET DESCRIPTION

3.1 Dataset Overview

The project uses a structured dataset of sales transactions. Each record represents one line item of an order, and the dataset contains approximately 9,800 records. The records belong to 4,922 distinct orders, 793 distinct customers, and 1,861 distinct products.

The dataset enables comparisons between customer segments and regions and the study of sales over time. The following table describes the columns used in the project.

3.2 Dataset Attributes

Table 3.1: Dataset Attributes Used in the Dashboard

Attribute	Description
Order ID	Identifier of the order; one order may contain several records
Order Date	Date on which the order was placed; used to derive the month
Customer ID	Identifier of the customer who placed the order
Segment	Customer segment: Consumer, Corporate, or Home Office
Region	Region of the order: West, East, Central, or South
Product ID	Identifier of the product sold
Sales	Sales value of the record
Month	Month number (1–12) derived from the order date

Note: The dataset may contain additional columns, such as shipping details, product category, and location. Only the attributes used in the dashboard are listed here.

3.3 Key Performance Metrics

Total Sales

Total Sales is the sum of the Sales column across all records.

$$\text{Total Sales} = \Sigma \text{ Sales}$$

Total Orders

Total Orders is the count of Order ID values in the dashboard, shown as 4.922K.

Total Customers

Total Customers is the number of distinct customers, shown as 793.

Total Products

Total Products is the number of distinct products sold, shown as 1.861K.

Derived Metrics

Average Sales per Record = Total Sales ÷ Number of Records ≈ 2.26M ÷ 9,800 ≈ 231

Average Sales per Order = Total Sales ÷ Total Orders ≈ 2.26M ÷ 4,922 ≈ 459

Average Sales per Customer = Total Sales ÷ Total Customers ≈ 2.26M ÷ 793 ≈ 2,850

These derived values are calculated from the rounded KPI values shown on the dashboard and are therefore approximate.

CHAPTER 4: METHODOLOGY

4.1 Project Workflow

The project follows these steps:

1. Dataset collection and loading.
2. Data inspection.
3. Data cleaning and preprocessing.
4. Exploratory data analysis using Python.
5. Calculation of key financial metrics.
6. Dashboard creation using Power BI.
7. Interpretation of analytical results.

Workflow:
Dataset → Data Cleaning → Data Analysis Using Python → Visualization Using Power BI → Insights and Conclusions

4.2 Data Loading

The CSV dataset is loaded into Python using the Pandas library. The dataset is inspected to understand its dimensions, column names, data types, and initial records. This helps identify the structure of the data before performing further operations.

4.3 Data Cleaning and Preprocessing

Data cleaning is an important step because missing or duplicate records and incorrect data types may affect the accuracy of the analysis.

The preprocessing stage includes:

- Checking the number of rows and columns.
- Inspecting column names and data types.
- Identifying missing values.
- Checking duplicate records.
- Converting the order date column into a date format.
- Extracting the month from the order date.
- Preparing the cleaned dataset for analysis.

Any missing values or duplicate records are handled according to the preprocessing logic used in the Python program.

4.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) is performed to understand the dataset and identify patterns.

The analysis includes:

- Calculating descriptive statistics for the Sales column.
- Counting unique orders, customers, and products.
- Grouping sales by customer segment.
- Grouping records by region.
- Aggregating sales by month.

The results help identify differences across segments and regions and variations across months.

4.5 Dashboard Development

The processed data is imported into Microsoft Power BI to develop a two-page dashboard. The dashboard uses KPI cards, a donut chart, a column chart, and a line chart. The first page focuses on customers, orders, and segment performance. The second page focuses on overall sales and the monthly trend.

CHAPTER 5: POWER BI DASHBOARD AND VISUALIZATIONS

5.1 Dashboard Overview

The Power BI report contains two pages. The first page presents customer and order metrics, the distribution of records by region, and sales by customer segment. The second page presents overall sales metrics and the monthly sales trend.

Together, these pages provide a visual summary of the financial performance recorded in the dataset.

5.2 Page 1: Customers, Orders and Segment Performance

Visualizations used:

- KPI card for total customers.
- KPI card for total orders.
- KPI card for average sales.
- Donut chart for customer records by region.
- Column chart for sum of sales by segment.

KPI Cards

Table 5.1: Page 1 KPI Values

KPI	Displayed Value
Total Customers	793
Total Orders	4.922K
Average Sales card	9.8K (see note)

Note: The third card is labeled Average Sales but is built on the Count of Sales field, so the value 9.8K represents the number of sales records and not an average amount. The average sales value per record is approximately 231 (2.26M ÷ 9,800).

Donut Chart: Count of Customer ID by Region

This chart shows how the 9.8K records are distributed across the four regions.

Table 5.2: Records by Region

Region	Records	Share
West	3.14K	32.04%
East	2.79K	28.42%
Central	2.28K	23.23%
South	1.6K	16.31%

Observation: The West region accounts for the largest share of records at about 32%, followed by East at about 28%. The South region has the smallest share at about 16%. The West and East regions together account for more than 60% of all records.

Column Chart: Sum of Sales by Segment

This chart compares total sales across the three customer segments. Based on the chart, the approximate values are:

Table 5.3: Approximate Sales by Segment

Segment	Approximate Sales	Approximate Share
Consumer	1.15M	51%
Corporate	0.69M	30%
Home Office	0.43M	19%

Note: These values are approximate readings from the chart and should be verified in Power BI before being included as exact results.

Observation: The Consumer segment contributes about half of total sales, roughly 1.7 times the Corporate segment and 2.7 times the Home Office segment.

5.3 Page 2: Overall Sales and Monthly Trend

Visualizations used:

- KPI card for total sales.
- KPI card for total orders.
- KPI card for total customers.
- KPI card for total products.
- Line chart showing the monthly sales trend.

KPI Cards

Table 5.4: Page 2 KPI Values

KPI	Displayed Value
Total Sales	2.26M
Total Orders	4.922K
Total Customers	793
Total Products	1.861K

Line Chart: Monthly Sales Trend

The horizontal axis shows the month number (1 to 12) and the vertical axis shows the sum of sales. Based on the chart, the approximate monthly sales are:

Table 5.5: Approximate Monthly Sales

Month	Approximate Sales	Month	Approximate Sales
1	95K	7	146K
2	60K	8	160K
3	198K	9	300K
4	137K	10	200K
5	155K	11	350K
6	146K	12	322K

Note: These values are approximate readings from the chart. Their total of about 2.27M agrees with the Total Sales KPI of 2.26M.

Observation: Sales are lowest in February at about 60K. A first peak appears in March, followed by a stable period from April to August at roughly 137K–160K per month. Sales then rise sharply in September, dip in October, and reach the highest point in November at about 350K. December remains high at about 322K.

Purpose:

- Identify seasonal patterns in sales.
- Identify the months with the highest and lowest revenue.
- Support planning for inventory, staffing, and promotions.

CHAPTER 6: RESULTS AND DISCUSSION

6.1 Results

The Financial Data Analytics project presents a two-page dashboard using Microsoft Power BI. The report summarizes key financial indicators and provides comparisons across customer segments, regions, and months.

The main findings from the displayed dashboard are:

Overall performance: The dataset records total sales of about 2.26 million across 4,922 orders, 793 customers, and 1,861 products.

Customer value: On average, each customer accounts for roughly 2,850 in sales and each order for roughly 459.

Segment comparison: The Consumer segment generates the highest sales (about 1.15M, or 51%), followed by Corporate (about 0.69M) and Home Office (about 0.43M).

Regional distribution: The West region has the largest share of records (32.04%), followed by East (28.42%), Central (23.23%), and South (16.31%).

Monthly trend: Sales are lowest in February and highest in November. The last four months of the year (September to December) contribute about 1.17M of the total, which is roughly half of annual sales.

Seasonality: The last quarter (October to December) alone contributes about 0.87M, or roughly 38% of total sales.

These findings are based on the values and patterns visible in the submitted dashboard screenshots.

6.2 Discussion

The dashboard shows that financial performance is concentrated in particular segments, regions, and months. Sales depend heavily on the Consumer segment and on the final months of the year, which suggests that promotions and stock planning should be focused on those areas.

The regional chart shows that the South region has the smallest share of records, which may indicate a market with room for growth or one that needs investigation.

The monthly trend shows that sales are strongly seasonal. The sharp rise from September and the peak in November are consistent with festive and year-end shopping periods, although the dataset alone does not confirm the reasons.

The analysis is descriptive and does not establish why a segment, region, or month performs better. Additional information, such as profit, discounts, product categories, and customer demographics, would be needed to investigate those factors.

CHAPTER 7: ADVANTAGES AND LIMITATIONS

7.1 Advantages

- Provides a visual summary of sales performance.
- Presents important financial metrics through KPI cards.
- Simplifies the comparison of segments and regions.
- Reveals seasonal patterns in sales.
- Combines Python-based preprocessing with Power BI visualization.
- Makes large transactional data easier to understand.

7.2 Limitations

- The analysis depends on the completeness and accuracy of the selected dataset.
- The dataset is a sample and may not represent real-time business performance.
- The dashboard shows only sales values and does not include profit, cost, or discount data.
- The regional chart counts records and does not show sales by region.
- The monthly trend combines data by month number, so year-over-year growth cannot be seen.
- The dashboard describes patterns but does not establish their causes.

7.3 Future Enhancements

The project can be improved in the following ways:

- Add profit, cost, and discount analysis.
- Show sales by region and by product category.
- Add year-over-year and month-over-month growth measures.
- Add interactive slicers for date, region, segment, and category.
- Identify the top customers and top-selling products.
- Use machine learning to forecast future sales.
- Connect the dashboard to regularly updated data.

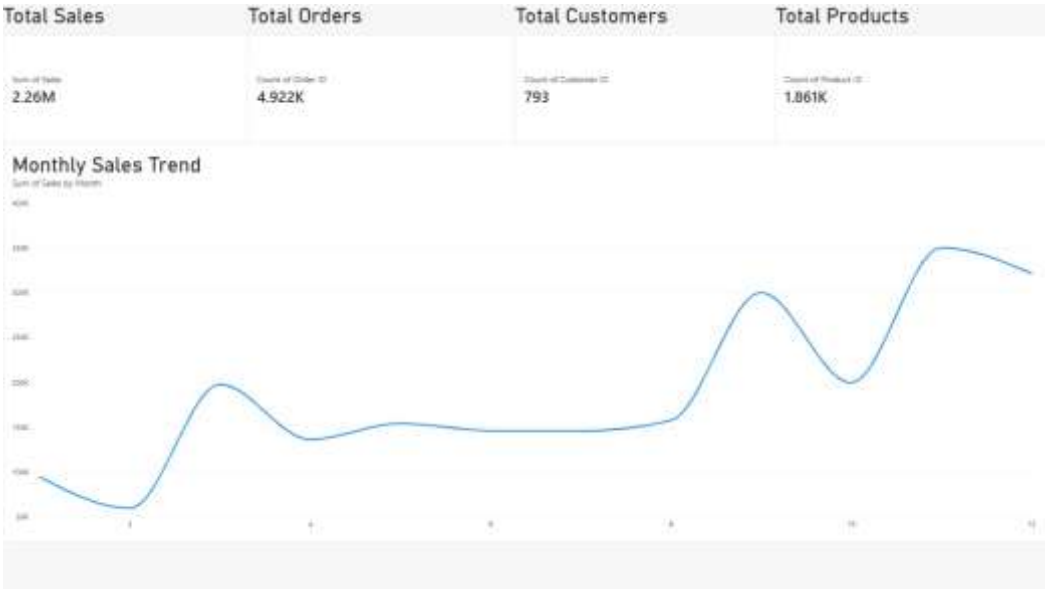
These enhancements could make the project more useful for financial analysts and business managers.

CHAPTER 8: POWER BI DASHBOARD OUTPUT

Figure 8.1: Page 1 – Customers, Orders and Segment Performance



Figure 8.2: Page 2 – Overall Sales and Monthly Trend



CHAPTER 9: CONCLUSION

The Financial Data Analytics project demonstrates the application of Python and Microsoft Power BI in analyzing and visualizing sales and financial data.

Python supports data inspection, preprocessing, and exploratory analysis, while Power BI presents the results through KPI cards and visualizations.

The dashboard shows total sales of about 2.26 million from 4,922 orders, 793 customers, and 1,861 products. It shows that the Consumer segment contributes the most sales, that the West region holds the largest share of records, and that sales peak in November with a strong year-end rise.

The analysis shows that financial performance varies by segment, region, and time period. Evaluating several dimensions together gives a more complete understanding of business performance than any single figure.

Overall, the project provides practical experience in data preprocessing, exploratory analysis, data visualization, and dashboard development. It demonstrates how raw transaction data can be converted into meaningful visual information that supports better planning and decision-making.